

Automated Cephalometric Landmark Localization Using Sparse Shape and Appearance Models

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Abstract. In this paper an automated method is presented for the localization of cephalometric landmarks in craniofacial cone-beam computed tomography images. This method makes use of a statistical sparse appearance and shape model obtained from training data. The sparse appearance model captures local image intensity patterns around each landmark. The sparse shape model, on the other hand, is constructed by embedding the landmarks in a graph. The edges of this graph represent pairwise spatial dependencies between landmarks, hence leading to a sparse shape model. The edges connecting different landmarks are defined in an automated way based on the intrinsic topology present in the training data. A maximum a posteriori approach is employed to obtain an energy function. To minimize this energy function, the problem is discretized by considering a finite set of candidate locations for each landmark, leading to a labeling problem. Using a leave-one-out approach on the training data the overall accuracy of the method is assessed. The mean and median error values for all landmarks are equal to 2.41 mm and 1.49 mm, respectively, demonstrating a clear improvement over previously published methods.

1 Introduction

Cephalometric analysis studies the craniofacial morphology and enables both orthodontists and maxillofacial, craniofacial or plastic surgeons to make a diagnosis, as well as to plan and evaluate an appropriate treatment. It consists traditionally of identifying predefined anatomical landmarks on lateral cephalometric radiographs. Based on these anatomical landmarks relevant distance and angular measurements are performed. A major disadvantage of this technique is its inherent two-dimensional character. The recent introduction of Cone Beam Computed Tomography (CBCT) in daily practice enables a truly three-dimensional cephalometric analysis on a routine basis, due to its low radiation dose, low cost and possibility of scanning the patient in its natural head position [1,2]. Since manual localization of the cephalometric landmarks in CBCT images is a

very tedious approach, there is a strong need for automated methods. However, automated localization of these landmarks is hampered by the large variability in craniofacial morphology, certainly in pathological cases, and the presence of artifacts in the images.

In this paper we present an automated method for the localization of cephalometric landmarks in CBCT images. This method makes use of a statistical sparse appearance and shape model. For a review of existing 2D automated methods, see [3]. In [4], Keustermans *et al.* present an automated method for the localization of cephalometric landmarks in 3D CBCT images. However, a major shortcoming is the manual definition of the connections between the cephalometric landmarks. Furthermore, a combination of a local and global shape model is used leading to an iterative algorithm.

This paper is organized as follows. Section 2 describes the proposed algorithm. Section 3 presents the model training phase of the algorithm, whereas Section 4 handles the model fitting. Subsequently, Section 5 discusses the experimental results. Finally, Section 6 formulates a conclusion and some ideas for future work.

2 Method

We present a supervised automatic landmark localization algorithm that incorporates prior information on these cephalometric landmarks. The use statistical models are trained on CBCT images with manually annotated landmarks. The prior information is twofold: first, the image intensity patterns in the CBCT image around each individual landmark leading to a statistical model describing appearance, and second, the spatial relationships between the landmarks leading to a statistical model describing shape.

The statistical model describing appearance is built for each cephalometric landmark individually, considering only a local region around the landmark in the CBCT image, hence leading to a sparse appearance model. Local image descriptors are used to describe the image intensity patterns in this local region. To build the statistical model describing shape, the landmarks are connected by edges, leading to a graph representation where the edges represent pairwise dependencies between landmarks. The edges are determined in an automated way based on the intrinsic graph topology present in the training data. Therefore, we search for dependencies among the cephalometric landmark positions by computing the description lengths of multivariate Gaussian models. By imposing the markovianity property onto the graph, each landmark is directly dependent upon its neighboring landmarks only, hence leading to a sparse shape model. The advantage of sparse shape models is the increased flexibility compared to global shape models. On the other hand, only pairwise dependencies can be modeled.

2.1 Bayesian Inference

The goal of the algorithm is the optimal localization of a number of predefined cephalometric landmarks. The accuracy of the solution can be expressed as a conditional probability on the location of the landmarks $\mathcal{L}$ given the image $\mathcal{I}$.

Using bayesian inference, this conditional probability can be reformulated as $P(\mathcal{L}|\mathcal{I}) = \frac{P(\mathcal{I}|\mathcal{L})P(\mathcal{L})}{P(\mathcal{I})}$. Here, $P(\mathcal{I}|\mathcal{L})$ is the image likelihood and $P(\mathcal{L})$ is the shape prior giving rise to, the sparse appearance model and the sparse shape model. For the remaining of this paper probabilities are converted into energies by taking the negative logarithm, leading to the following energy function to be minimized

$$\mathcal{L}^* = \arg \min_{\mathcal{L}} (E_I(\mathcal{I}, \mathcal{L}) + E_S(\mathcal{L})) \quad , \quad (1)$$

where $E_I(\mathcal{I}, \mathcal{L})$ and $E_S(\mathcal{L})$ are the negative logarithm of $P(\mathcal{I}|\mathcal{L})$ and $P(\mathcal{L})$, respectively.

2.2 Sparse Appearance Model

The image likelihood is simplified by making two assumptions. First, we assume that the influence of the presence of a landmark on the image is only local, justifying the use of local image descriptors to capture the local intensity patterns around each landmark. Second, we assume that local image descriptors are mutually independent for each landmark. In this paper, the nSIFT image descriptor is used [5]. This descriptor consists of a set of gradient histograms over a window centered on each landmark.

Using the aforementioned assumptions the image likelihood term can be rewritten as follows

$$P(\mathcal{I}|\mathcal{L}) = \prod_i^n P(\mathcal{I}|l_i) = \prod_i^n P_i(\omega_i) \quad . \quad (2)$$

In this equation l_i represents the i^{th} anatomical landmark, n is the number of landmarks and ω_i represents the local image descriptor corresponding to landmark l_i . Finally, converting the probabilities into energies by taking the negative logarithm yields the following expression:

$$E_I(\mathcal{I}, \mathcal{L}) = \sum_i^n d_i(\omega_i) \quad , \quad (3)$$

where d_i is the energy function corresponding to landmark i .

2.3 Sparse Shape Model

The shape prior introduces the sparse shape model into the Bayesian framework. Two assumptions are made as well. First, invariance of the shape model with respect to translations is assumed. Second, we assume that a landmark directly interacts with its neighbors only, thereby imposing the sparse nature of the shape model. The sparse shape model is constructed by embedding the landmarks in an undirected graph $\mathcal{G}$. The undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ consists of a set of nodes $\mathcal{V} = \{l_1, \dots, l_n\}$, the landmarks, and a set of edges $\mathcal{E}$. Let $\mathcal{N}$ be a neighborhood

system defined on $\mathcal{V}$, where $\mathcal{N}_i = \{l_j \in \mathcal{V} | (l_i, l_j) \in \mathcal{E}\}$ denotes the set of neighbors of l_i . A clique c of the graph $\mathcal{G}$ is a fully connected subset of $\mathcal{V}$. The set of cliques is represented by $\mathcal{C}$. For each $l_i \in \mathcal{V}$, let X_i be a random variable taking values x_i in some discrete or continuous sample space $\mathcal{X}$. By concatenating the variables at each node, we obtain a random vector $\mathbf{X} = \{X_i | l_i \in \mathcal{V}\}$. By the Hammersly-Clifford theorem a Markov Random Field can be defined in terms of a decomposition of the distribution over cliques of the graph:

$$p(\mathbf{x}) = \frac{1}{Z} \exp \left(- \sum_{c \in \mathcal{C}} V_c(\mathbf{x}) \right) , \quad (4)$$

where Z represents the partition function and V_c represents the clique potential function. In the algorithm presented in this paper only pairwise cliques are used. Converting probabilities into energies and taking into account the translation invariance leads to the following energy function for the local shape model

$$E_S(\mathcal{L}) = \sum_{(i,j) \in \mathcal{E}} d_{ij}(\mathbf{x}_j - \mathbf{x}_i) , \quad (5)$$

where d_{ij} is the energy function of the edge between the landmarks l_i and l_j and corresponds to the clique potential function.

2.4 Edge Definition

The pairwise connections or edges between the cephalometric landmarks encode spatial dependencies in the sparse shape model. To make the sparse shape model as robust as possible, only sensible edges should be defined. This is, however, a complex task, highly dependent upon the training data. To analyze the intrinsic spatial dependencies between the cephalometric landmarks in the training data, we compute the description length of a multivariate Gaussian model for each pairwise combination of landmarks. The description length quantifies the compactness of the model capturing the edge variation. The rationale behind the compactness is the assumption that a high spatial dependency between two landmarks leads to a more compact model. The description length comprises the cost (number of bits) of communicating a model itself and the data encoded with the model. For more details we refer to Davies *et al.* [7]. In this framework we use a simplified version of the description length as presented by Thodberg [8]. This approach is similar to the method presented by Langs *et al.* [9]. In this paper, however, only pairwise subsets are used and no alignment of the subsets is performed. The use of a multivariate Gaussian edge model constrains the pairwise dependencies to be linear. However, this approach can be extended to nonlinear dependencies as well, using kernel principal component analysis [6] (see Section 3).

Figure 1 gives the result of applying this method to the training data of the cephalometric landmarks. From this figure it can be seen that the left-right

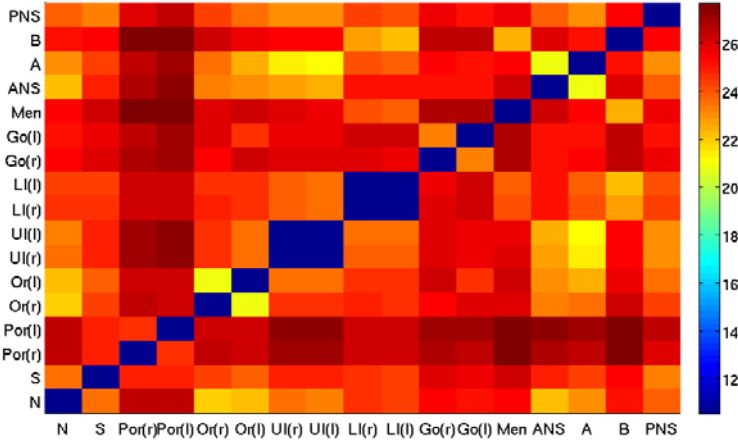


Fig. 1. Description length for all pairwise connections between the cephalometric landmarks. Blue indicates a description length of value 10 or lower, whereas red indicates a description length of value 26 or higher.

connections between the landmarks have a low description length (*upper incisor* (UI), *lower incisor* (LI), *orbitale* (Or) and *gonion* (Go)). Furthermore, the description length of any connection to the *porion* (Por) is high. This is caused by artifacts present in the CBCT at the level of this landmark, hampering the correct localization of this landmark.

Given the description length for all pairwise connections, edges need to be defined. Therefore we apply a weight to each edge, depending on its description length. An edge with a low description length will be assigned a high weight, whereas an edge with a high description length will be assigned a low weight, making it less important in the sparse shape model. For the assignment of the weights, the sigmoid function is used.

3 Model Training

For the sparse appearance and shape model, probability density functions need to be estimated from the training data. For the sparse shape model a Gaussian distribution provides good results. For the sparse appearance model, on the other hand, a more advanced method is needed, which is related to kernel principal component analysis [6]. It consists of a nonlinear mapping $\phi : \mathbb{R}^d \mapsto F$ of the training samples, which are the nSIFT feature descriptors in each landmark, to a higher (possibly infinite) dimensional feature space F in which a Gaussian distribution is presumed. There is no need to explicitly compute the nonlinear mapping ϕ , since only scalar products in feature space are needed that, by use of the Mercer theorem, can be evaluated using a positive definite kernel function $k(\omega_i, \omega_j) = \langle \phi(\omega_i), \phi(\omega_j) \rangle$. The Gaussian kernel is used with a bandwidth equal to the mean nearest-neighbor training sample distance.

4 Model Fitting

The actual localization of the landmarks in a CBCT image comes down to optimizing the energy function (1). This is not a straightforward operation since the energy function is not convex. However, due to the recent advances in solving the inference problem in multilabel Markov Random Fields the non-convexity can be partially solved by a discretization of the energy function.

4.1 Discretization

The discretization of the energy function is performed by imposing a discrete sample space $\mathcal{X}$ on the graph $\mathcal{G}$. This discrete sample space consists of a finite set of candidate landmark locations. As such, the optimization problem comes down to the selection of the most optimal candidate location for each landmark. These candidate locations are obtained by evaluating the image likelihood $d_i(\omega_i)$ in a search grid located around the expected location of the landmark of interest and selecting the m locations with lowest cost. This results in a set of candidates $\mathbf{x}_i = \{\mathbf{x}_{ik}\}_{k=1}^m$ for each landmark. Important to note is that we assume that the CBCT image is rigidly registered to the images in the training data set. The optimization problem now becomes a labeling problem: $\mathbf{r} = \{\mathbf{r}_i \in \{0, 1\}^m\}$. Following conditions must hold: $\sum_{k=1}^m r_{ik} = 1$ and $r_{ik} = 1$ if candidate k is selected. The resulting discrete cost function to be minimized becomes

$$\arg \min_{\mathbf{r}} \left(\sum_{i=1}^n \sum_{k=1}^m r_{ik} d_i(\omega_{ik}) + \gamma \sum_{i=1}^n \sum_{k=1}^m \left(r_{ik} \sum_{j=1}^n \sum_{o=1}^m r_{jo} d_{ij}(\mathbf{x}_{ik}, \mathbf{x}_{jo}) \right) \right), \quad (6)$$

where γ is a constant that determines the relative weight of the image likelihood and the shape prior.

4.2 Markov Random Field Optimization

Currently two types of methods are most prominent in multilabel Markov random field optimization: the methods based on graph cuts [10] and those based on message-passing [11,12]. However, the methods based on graph cuts can not be used to minimize the energy function (6) because it is not submodular [13]. Here a variant of the Belief Propagation (BP) algorithm is used. Belief propagation is a local message passing algorithm that converges to a fixed point for an acyclic graph. Since the graphs obtained by the method in this paper in general do contain cycles, we use the BP-fusion algorithm of Lempitsky *et al.* [14]. This BP-fusion algorithm combines different belief propagation proposals using the fusion move algorithm.

5 Experiments and Results

The algorithm described in this paper is used to identify 17 cephalometric landmarks in CBCT images. These landmarks are *nasion*, *sella*, *porion* (left and

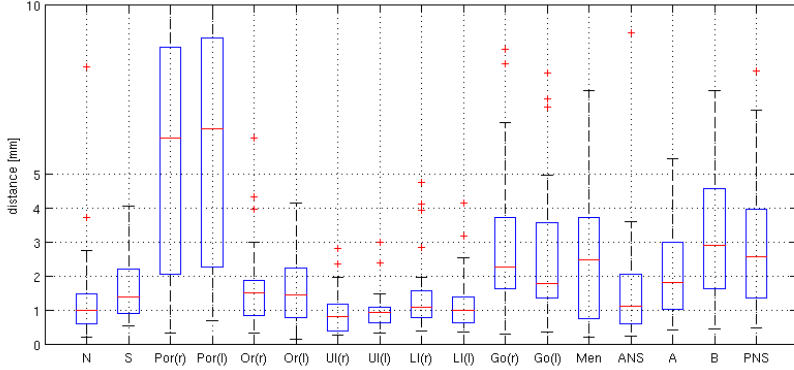


Fig. 2. Box plot of the errors for each landmark using a leave-one-out approach on the training data

right), *orbitale* (left and right), *upper incisor* (left and right), *lower incisor* (left and right), *gonion* (left and right), *menton*, *anterior nasal spine*, *A-point*, *B-point* and *posterior nasal spine* [1]. The training data set consisted of 37 CBCT images ($0.4 \times 0.4 \times 0.4$ mm) from patients undergoing maxillofacial surgery, with manually identified landmarks. A leave-one-out approach is used to validate the algorithm.

Figure 2 shows the results of the leave-one-out approach. The mean and median error values for all landmarks are equal to 2.41 mm and 1.49 mm respectively. Compared to the results obtained by Keustermans *et al.* [4], there is an improvement of 0.14 mm and 0.23 mm for the mean and median error respectively. The median error values separated in the transversal, sagittal and transversal direction are equal to 0.58 mm, 0.55 mm and 0.61 mm, respectively. This allows to compare the results to the inter- and intra-observer variability of manual landmark localization, which is equal to respectively 0.78 mm, 0.86 mm and 1.26 mm for the inter-observer variability and 0.88 mm, 0.76 mm and 0.84 mm for the intra-observer variability, as reported in [1].

6 Conclusion

A statistical model-based algorithm for the localization of cephalometric landmarks in CBCT images incorporating both a sparse appearance and shape model is presented. The sparse appearance model is constructed through the use of local image descriptors. The sparse shape model is obtained by embedding the landmarks in a graph. The edges of the graph are defined based on the intrinsic dependencies between the landmark positions present in the training data. The energy function, obtained from a maximum a posteriori approach, is optimized efficiently using multilabel Markov Random Fields. The accuracy of this method is shown using a leave-one-out approach on the training data, demonstrating an

outperformance over existing methods. Since the training data are obtained from patients undergoing maxillofacial surgery, the robustness of the algorithm with respect to pathologies is shown.

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